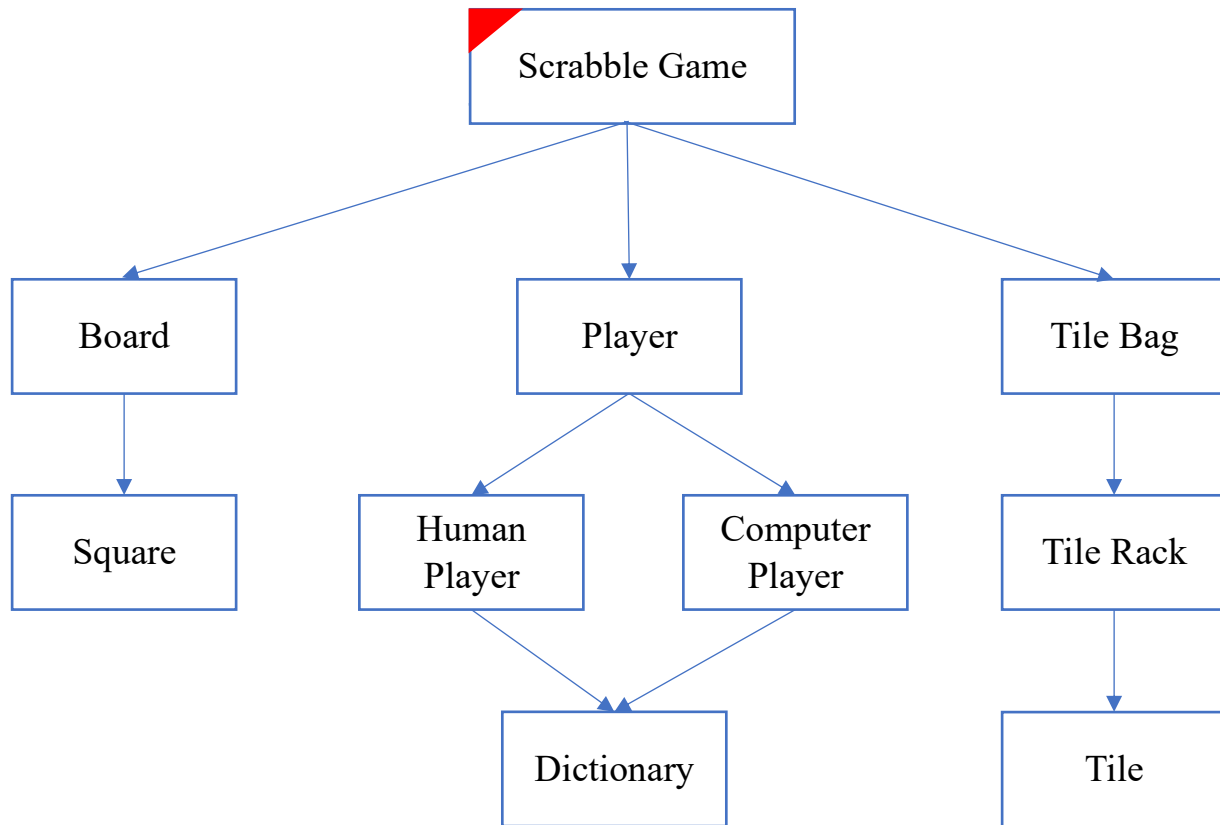


Proposed Design



Trigger: update from mouse

Description of the Design

- Scrabble Game

Data: Board, Human and Computer players, Tile Bag instance, Dictionary, Player Tracker,

Behavior: Initialize game components, manage game flow and turns, validates and process moves, handle tile exchanges, track game progress, calculates and updates scores, determines game end conditions

- Board

Data: Grid of squares, premium square positions map, first move state

Behavior: Initialize board layout, place and remove tiles, validate tile placements, check first move status, provide square information, calculate word scores, track premium square usage

- Player

Data: Player name, score, tile rack, player type

Behavior: Make moves, exchange tiles, draw new tiles, update score, manage tile rack

- Human Player

Data: Player's name, current tiles, current score.

Behavior: Move management, validation of the placement, finds all the words created, finds word at position

- Computer Player

Data: Player's name, current tiles, current score.

Behavior: Move management, finds playable words, check word possibilities, find valid positions, identifies cross words, validates move, check connections, verifies word validity, calculates total score, scores cross words, assess move value, gets letter score, finds letter position, exchanges tiles, chooses tiles to swap.

- Tile Bag

Data: Collection of remaining tiles

Behavior: Initialize tile distribution, draw random tiles, return exchanged tiles, shuffle tiles, track tile count.

- Dictionary

Data: List of valid words

Behavior: Load dictionary file, validate words, provide word existence check.

- Square

Data: Position on board, current tile (if any), square type (normal, double letter, etc.)

Behavior: Hold a tile, provide score multipliers

- Player

Data: Name, score, current rack of tiles

Behavior: Play moves, exchange tiles, pass turns

- Tile

Data: Letter value, point value, blank status, current letter

Behavior: Provide letter value, provide point value, handle blank tile assignment, check blank status